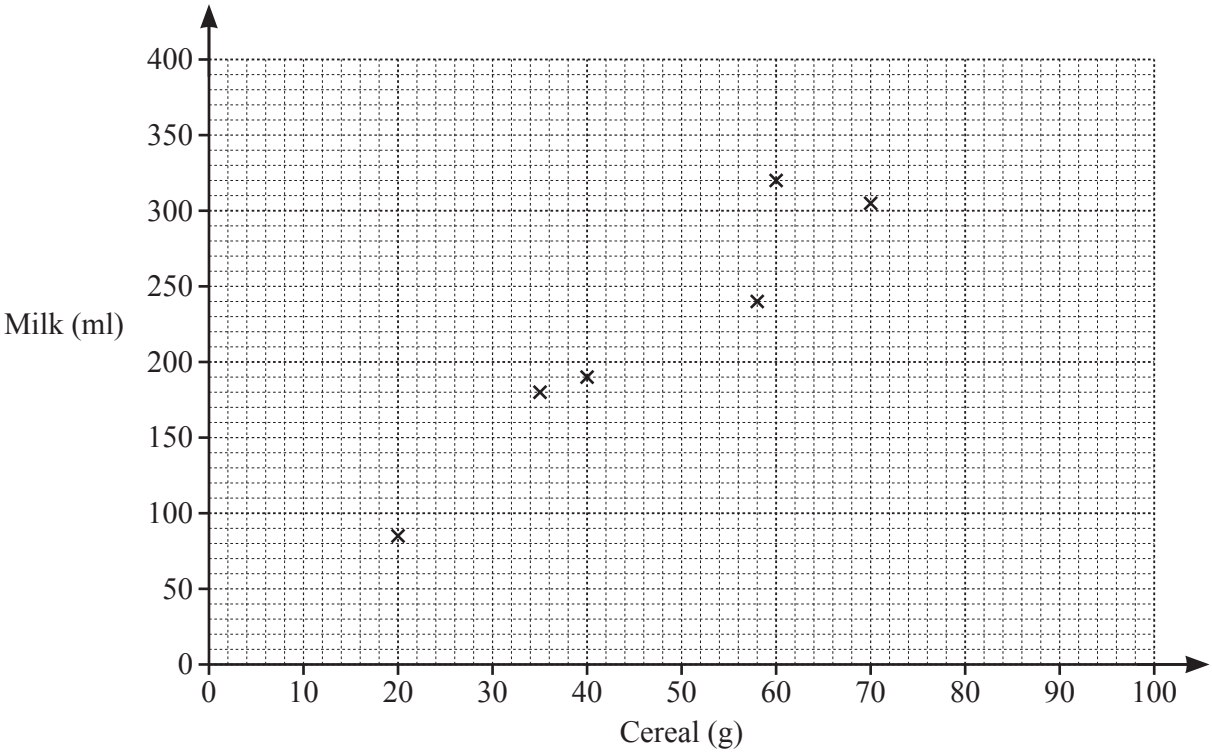
0580_s20_qp_33_Q6

6 **(a)** Ten students eat cereal with milk for breakfast.
The amounts are shown in the table.

Cereal (g)	40	20	58	70	60	35	28	40	55	46
Milk (ml)	190	85	240	305	320	180	150	230	340	220



(i) Complete the scatter diagram.
The first six points have been plotted for you. [2]

(ii) For these students, describe the relationship between the amount of cereal and the amount of milk.

..... [1]

(iii) On the grid, draw a line of best fit. [1]

(iv) Another student has 280 ml of milk with her cereal.

Use your line of best fit to estimate an amount of cereal this student has.

..... g [1]

(v) Explain why this scatter diagram should not be used to estimate the amount of milk for a student who has more than 70 g of cereal.

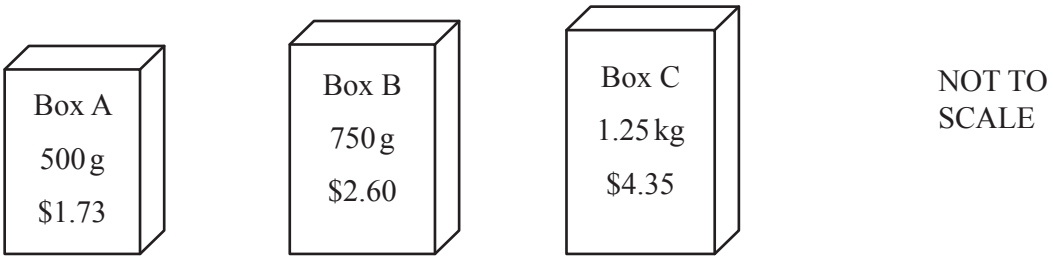
..... [1]

(b) 100 g of cereal contains 360 kilocalories.
100 ml of milk contains 45 kilocalories.
For breakfast Sasha has 35 g of cereal with 180 ml of milk.

Work out the number of kilocalories Sasha has for breakfast.

..... kcal [3]

(c) A shop sells cereal in boxes A, B and C.



Work out which box is the best value.
You must show all your working.

Box ..... [3]